

Woo-Seong Yun

ML ENGINEER | AI RESEARCHER

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Summary

Hi! I'm Woo-Seong Yun, a Machine Learning Engineer at Watcha. My research began from a question: can AI discover preferences that users haven't yet recognized themselves? I have tackled fundamental challenges in recommender systems and large language models, from popularity bias and cold-start to model interpretability. I have published papers on LLMs (ICML 26), LLM-based Recommendation (WSDM 26), and Recommender Systems (CIKM 26, RecSys 25, Applied Soft Computing), etc. I enjoy exploring innovative solutions to fundamental problems and continually expanding my expertise through new techniques and tools

Experience

WATCHA Inc.

Dec. 2025 - Present

MACHINE LEARNING ENGINEER

- Recommender systems for Watcha's OTT & content-rating services.
 - GD-SSM (CIKM'26): Mitigates popularity bias via gradient decoupling in production recommenders.
 - Cold-start Onboarding: Preference Elicitation for cold-users
 - Explainable Recommendation with Negative Feedback
 - Cross-domain Sequential Recommendation across content domains
- Offline evaluation and online A/B test design & analysis for model iteration.

Chung-Ang University

Sep. 2025 - Nov. 2025

AI RESEARCHER

- ARROW (WSDM'26): Adaptive reasoning for LLM-based recommendation with explainability
- LINKER: Leveraging modality knowledge for semantic relation generation in multimodal recommendation

Education

Chung-Ang University

Sep. 2023 - Aug. 2025

M.S. IN ARTIFICIAL INTELLIGENCE

- GPA: 4.34/4.50
- Data Science Lab, advised by Prof. Yoon-Sik Cho.
- Master's thesis: Two Contrastive Approaches for Addressing the Long-Tail Problem in Recommender Systems.

Chung-Ang University

Mar. 2018 - Aug. 2023

B.S. IN APPLIED STATISTICS

- GPA: 4.20/4.50
- Summa Cum Laude, Early Graduation

Publications

[C1] Norm Removed, Bias Remains: Gradient-Decoupled Sampled Softmax for Collaborative Filtering

CIKM 2026

WOO-SEONG YUN (WATCHA INC.)*, YEOJUN CHOI*, AND YOON-SIK CHO

[C2] Capacity without Access: Reinterpreting the Mid-Depth Spectral Plateau in LLMs

ICML 2026

SEONG-MIN KANG, WOO-SEONG YUN, NA-HYUN LEE, AND YOON-SIK CHO

[C3] ARROW: Adaptive Reasoning for LLM-based Recommendation with Explainability

WSDM 2026

WOO-SEONG YUN, MIN-SEONG KIM, AND YOON-SIK CHO

[C4] Rethinking Overconfidence in VAEs: Can Label Smoothing Help?

RecSys 2025

WOO-SEONG YUN*, YEOJUN CHOI*, AND YOON-SIK CHO

[C5] Relational Alignment Graph Neural Network for Multimodal Recommendation

IEIE 2025

WOO-SEONG YUN, MYUNGBIN KWAK, EUNSUN KIM, AND YOON-SIK CHO

[C6] Variational Autoencoder with Label Smoothing for Collaborative Filtering

KIIT 2024

WOO-SEONG YUN, YEONGHYEON KIM, CHANWOO JEONG, YEOJUN CHOI, AND YOON-SIK CHO

[J1] Contrastive Disentangled Variational Autoencoder for Collaborative Filtering

IEEE Access

WOO-SEONG YUN, SEONG-MIN KANG AND YOON-SIK CHO

[J2] GCNs Meet Long-Tail: Embedding Norm Bias in GCN-Based Recommendations

Applied Soft Computing

YEOJUN CHOI, WOO-SEONG YUN, CHANWOO JEONG, AND YOON-SIK CHO

[J3] Enhancing LLMs for Sequential Recommendation with Reversed User History and User Embeddings

IEEE Access

YEOJUN CHOI, WOO-SEONG YUN, AND YOON-SIK CHO

Preprints

[P1] All Items are not Equal: Addressing the Imbalance of Item Knowledge in LLM for Recommendation

under review

WOO-SEONG YUN*, YEONGHYEON KIM*, AND YOON-SIK CHO

[P2] LINKER: Leveraging Modality Knowledge for Semantic Relation Generation in Multimodal Recommendation

under review

MYUNGBIN KWAK*, WOO-SEONG YUN*, AND YOON-SIK CHO

Projects

LG HE PROJECT

CORPORATE ASSIGNMENT PROJECT

Jul. 2024 - Dec.2024

- Development of a 'sequence-aware recommender system' using TV watching data.
- Collaborated with Hanyang University DCX Lab as and participate as the **Project Manager** at Chung-Ang University.
- Built sequential models (SASRec, Bert4Rec) with sliding-window and time-interval-aware augmentation
- Achieved Recall@10 4.9× over a popularity baseline (0.132 → 0.650).

Hyundai Glovis Project

CORPORATE ASSIGNMENT PROJECT

Apr. 2023 - Dec.2023

- Development of an AI model for recommending packaging specifications and automating MH calculation.
- Built a packaging recommendation model using an ensemble of data matching and similarity models.
- Identified key factors and examined the impact of each variable through feature analysis.
- Achieved 95% prediction accuracy.

Restaurant Recommendation Chatbot

HANIUM AI PROJECT, SHINHAN CARD Co.

Mar. 2022 - Dec.2022

- Development of a system that recommends restaurants to consumers with diverse social values.
- Implementation of a rule base recommender system based on various scenarios using Dialogflow.
- Built and deployed a full Python pipeline (Dialogflow, KakaoTalk, database) as a live service.

Awards & Scholarships

2025 **Gold Prize for Best Paper Award in Special Session of IEIE**, Chung-Ang University AI Graduate School

2023-2025 **CAU GRS Scholarship**, Chung-Ang Graduate School

2023 **Summa Cum Laude Honor**, Chung-Ang University

2018 **Academic Excellence Scholarship**, Chung-Ang University

Extracurricular Activities

Undergraduate Research

DATA SCIENCE LAB (PROF. YOON-SIK CHO)

- Participation in the Hyundai Glovis Project – Packaging specification recommendation team.
- Assistance in research experiments on state-of-the-art recommender systems utilizing Deep Neural Networks.

Seoul, Korea

Dec. 2022 - Jul. 2023

COSADAMA

CORE MEMBER

- Participation in a project analyzing factors influencing Korean Stock Prices.
- Assistance in Elementary Statistics and Python Course.
- Completed Data Science, Machine Learning, SQL, Web Crawling Course.

Seoul, Korea

Mar. 2022 - Dec. 2022

NK Statistical Programming

NAMKANG HIGH SCHOOL TUTOR

- Designed and conducted an R-based statistical programming course for high school students.
- Assistant for students' capstone projects.

Seoul, Korea

Jul. 2021 - Jul. 2023

Qualifications

ADsP

Word Processor

Korean History Proficiency Test : Level 1